Technical Documentation

Legal Document RAG Assistant

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1 Introduction

1.1 What is RAG

Retrieval-Augmented Generation (RAG) combines:

- Information retrieval
- Language generation

1.2 Why RAG

- Reduces hallucination
- Improves accuracy

1.3 Use Case

Legal document analysis and question answering.

2 System Architecture Explanation

Documents → Embeddings → Vector Store
Query → Retrieval → LLM → Answer
LangGraph controls workflow execution

3 Design Decisions

3.1 Chunk Size

500 with overlap 100. Balances context and performance.

3.2 Embedding Strategy

MiniLM used for fast computation.

3.3 Retrieval Approach

Top-k similarity search for relevant chunk selection.

3.4 Prompt Design

- No hallucination
- Context-only answers

4 Workflow Explanation

4.1 LangGraph Control

LangGraph manages:

- Routing
- Processing
- Escalation

4.2 Node Responsibilities

- route_node → decision
- process_node → answer
- hitl_node → escalation

5 Conditional Logic

5.1 Intent Detection

Simple rule-based logic is used.

5.2 Routing Decisions

- Answer → if confident
- Escalate → otherwise

6 HITL Implementation

6.1 Role

Handles complex queries.

6.2 Benefits

Improves reliability.

6.3 Limitations

Requires human availability.

7 Challenges & Trade-offs

7.1 Retrieval Accuracy vs Speed

Higher k improves context but increases latency.

7.2 Chunk Size vs Quality

Larger chunks improve context but slow processing.

7.3 Cost vs Performance

API-based LLM provides better performance but incurs cost.

8 Testing Strategy

8.1 Approach

Test with multiple legal queries.

8.2 Sample Queries

- “What is the termination clause?”
- “What are the penalties?”

9 Future Enhancements

- Multi-document support
- Feedback loop
- Memory integration
- Web-based UI
- Cloud deployment